Electrical design in progress; see current PCB and check reports for completion status.

Review items: actual battery; clock loads; USB mechanical drawing and inrush.

KiCad10.0.7 is available. ERC/DRC status is recorded in reports, not inferred from this file.

Use README.md and docs/release_checklist.md before continuing design.

POWER PATH / BATTERY

Plik: power.kicad_sch

MCU / RADIO / CLOCK

Plik: mcu.kicad_sch

IMU / MEMORY

Plik: sensors.kicad_sch

USB / CONTROLS / SWD

Plik: io.kicad_sch

BATTERY TEMPERATURE

Plik: temperature.kicad_sch

NOT FOR FABRICATION – SEE CHECK REPORTS

Sheet: /
File: Ankle_Sensor_V1.kicad_sch

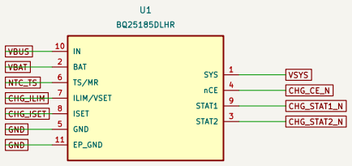
Title: Ankle Sensor V1 – WORK IN PROGRESS

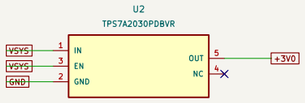
Size: A3	Date: 2026-10-28	**Rev: A1-WIP**
KiCad E.D.A. 10.0.7		Id: 1/6

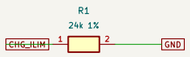
POWER PATH / BATTERY — REVIEW DRAFT

4.2V protected1S pack only. 100mA USB input limit. 150mA ISET is dynamically reduced.

See temperature sheet for independent NTC window. Pack/NTC qualification is mandatory.



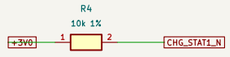




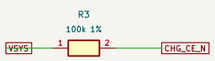




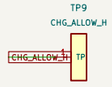






















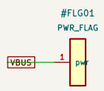


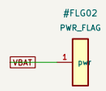


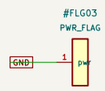














WORK IN PROGRESS — SEE CURRENT ERC/DRC REPORTS

Sheet: /POWER PATH / BATTERY/
File: power.kicad_sch

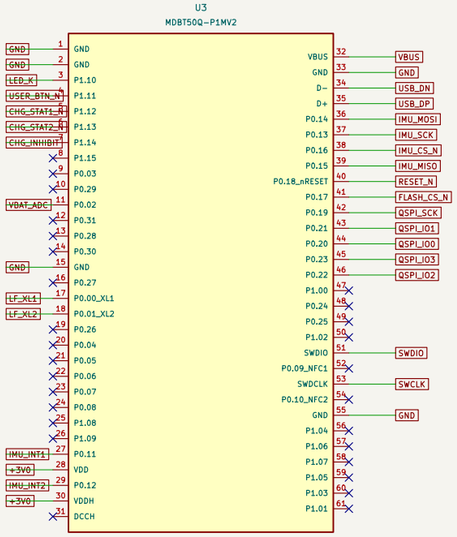
Title: Ankle Sensor V1 / power

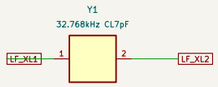
Size: A3	Date: 2026-10-28	**Rev: A1-WIP**
KiCad E.D.A. 10.0.7		Id: 2/6

MCU / RADIO / CLOCK – REVIEW DRAFT

Normal voltage: VDD=VDDH=3V0. DCCH reference wiring is a release blocker.

QSPI/SPI on fast P0 GPIO. See pinmap.csv. No antenna geometry has been guessed.







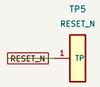












LFXO loads and USB bypass capacitors are provisional pending reference circuit review.

WORK IN PROGRESS – SEE CURRENT ERC/DRC REPORTS

Sheet: /MCU / RADIO / CLOCK/
File: mcu.kicad_sch

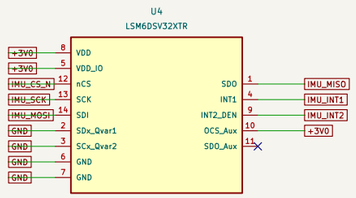
Title: Ankle Sensor V1 / mcu

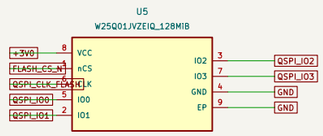
Size: A3 | Date: 2026-10-28 | **Rev: A1-WIP**
KiCad E.D.A. 10.0.7 | Id: 3/6

IMU / NOR FLASH — REVIEW DRAFT

SPI8MHz; INT1 FIFO watermark; INT2 can serve timestamp/ODR-trigger experiments.

U5 upgraded to128MiB, same8x6mm body. Address32bit and dual-die status handling.





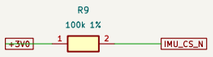


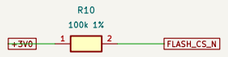


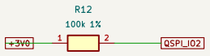


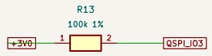






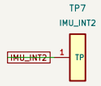












WORK IN PROGRESS — SEE CURRENT ERC/DRC REPORTS

Sheet: /IMU / MEMORY/
File: sensors.kicad_sch

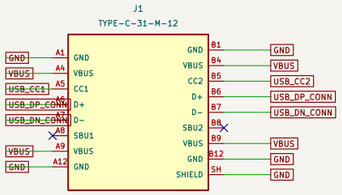
Title: Ankle Sensor V1 / sensors

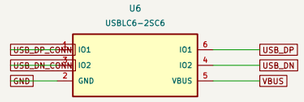
Size: A3	Date: 2026-10-28	**Rev: A1-WIP**
KiCad E.D.A. 10.0.7		Id: 4/6

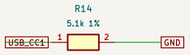
USB2 / USER INTERFACE / SWD — REVIEW DRAFT

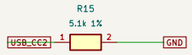
Two independent Rd5.1k. USB optional data path is physically connected.

USB contact-to-footprint mapping, CC/VBUS protection and inrush analysis remain open.







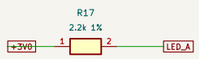


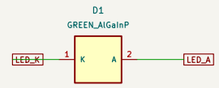


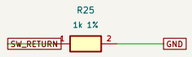












SWD pogo pin numbering is project-specific; VTref is not a power input.

WORK IN PROGRESS — SEE CURRENT ERC/DRC REPORTS

Sheet: /USB / CONTROLS / SWD/
File: io.kicad_sch

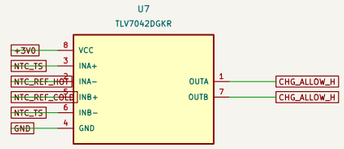
Title: Ankle Sensor V1 / io

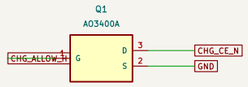
Size: A3	Date: 2026-10-28	**Rev: A1-WIP**
KiCad E.D.A. 10.0.7		Id: 5/6

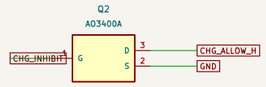
INDEPENDENT BATTERY NTC WINDOW — REVIEW DRAFT

NTC103AT-2 /10k B3435 must be on CELL. No10k bypass resistor is permitted.

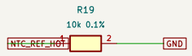
Firmware P1.14 HIGH=inhibit viaQ2; LOW/HIZ permits hardware control.

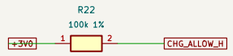




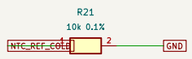


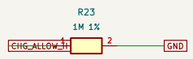




















U8 holdsALLOWlow during reference startup. See tolerance calculation and tests.

Pack acceptance: ROC>=24k, R45C<=5.20k, monotonic. Physical qualification required.

WORK IN PROGRESS — SEE CURRENT ERC/DRC REPORTS

Sheet: /BATTERY TEMPERATURE/
File: temperature.kicad_sch

Title: Ankle Sensor V1 / temperature

Size: A3	Date: 2026-10-28	**Rev: A1-WIP**
KiCad E.D.A. 10.0.7		Id: 6/6